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**Accelerating Biodiesel Catalytic Production by Confined Activation
of Methanol over High-Concentration Ionic Liquids Grafted UiO-66
Solid Superacids**

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KEYWORDS: Metal-organic frameworks; solid acids; heterogeneous catalysis; biomass conversions; superacidity.

ABSTRACT

Solid acids usually show lower catalytic activities than liquid acids due to their limited acid strength and presence of obvious diffusion limitation in the reactions. We report herein a variety of ionic liquids grafted UiO-66 solid superacids synthesized from quaternization of UiO-66 with 1,3-propane sultone, ion-exchanging with H_2SO_4 or HSO_3CF_3 . The developed UiO-66 solid superacids are high-efficient in catalyzing biodiesel production towards transesterification, which were better than the ionic liquid and neat H_2SO_4 . The remarkable activities found in the UiO-66 solid superacids benefit from the synergistic effect of high acid concentrations (~ 3.33 mmol/g), superacidity (determined by ^{31}P solid-state NMR, Hammett indicators, and potentiometric titration), as well as unique shape-selective confinement for methanol into the ordered and uniform micropores, which targeted activation of methanol to prevent the occurring of inverse reaction in the reactions. The UiO-66 solid superacids show important application potentials in the area of biodiesel production in the industry.

INTRODUCTION

As one class of *state-of-the-art* nanomaterials, metal-organic frameworks (MOFs) have received considerable research interests due to their unique structural features such as crystalline frameworks, high porosity, huge internal surface areas, tunable pore sizes and controllable metal-organic hybrid building units^{1–5}. MOFs have been widely applied in the areas of gas adsorption and separation^{6–11}, drug delivery^{12,13}, energy storage and conversion^{14–16}, as well as heterogeneous catalysis^{17–20}. Heterogeneous catalysis was one of the earliest proposed applications for the MOFs nanomaterials²¹, MOFs show great significant feature of their potential inner porosity that facilitate the mass transportation of guest molecules and accessibility of catalytically active sites in the micropores, which largely improve their catalytic activities in various reactions^{17–21}. Recently, numerous MOFs derived solid catalysts such as MOFs with noble metals^{22,23}, reactive functional groups^{24–26}, acid and base active sites^{27–29} have been successfully developed. The synthesized MOFs solid catalysts show superior activities in different kinds of reactions such as selective oxidation, asymmetric catalysis, photocatalysis, electrocatalysis, and acid-base catalysis^{21–23,24–26}.

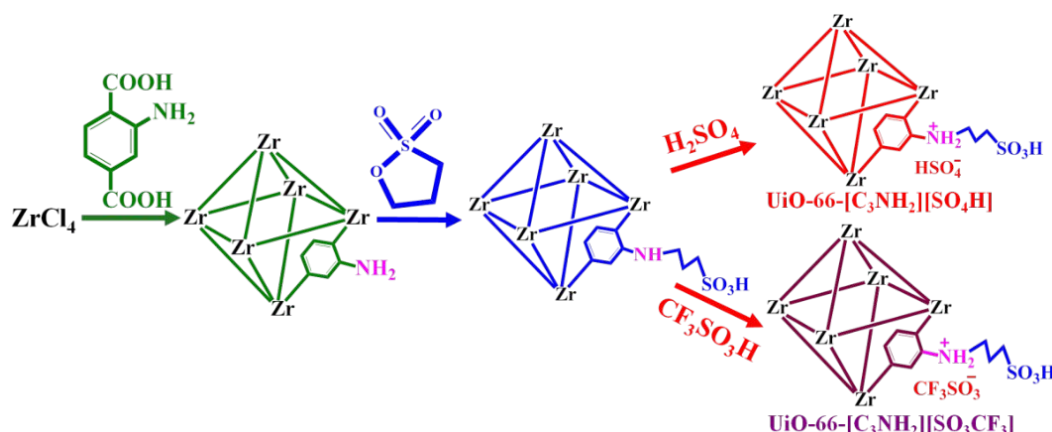
As the well-known heterogeneous catalysts, solid acids have been extensively studied for developing green and sustainable chemistry due to their good catalytic activities and products selectivities, easy regeneration, reduced corrosion, and can be easily operated in continuous reactors^{30–32}. MOFs can be used as the promising candidates for designing of new kinds of solid acids due to their following

advantages: i) the metal centers simultaneously acts as the structural building unit and Lewis acid sites for catalytic applications^{26,33,34}; ii) ultraporous structures and ordered microporosity result in the solid acids with improved mass transportation and unique confinement properties in comparison with other kinds of porous supports²; iii) the tunable organic-metal building units make the MOFs catalysts show good wettability for various reactants, which strongly enhances their compatibility in various reaction systems⁴. To date, the widely used strategy for preparation of MOFs solid acids was to graft Brønsted acid sites such as $-\text{SO}_3\text{H}$ onto their frameworks, which can be achieved from sulfonation reaction^{27,34-36}. The synthesized MOFs solid acids have been widely applied in many acid-catalyzed reactions such as Friedel-Crafts acylation, esterification, isomerization and biomass conversion^{27, 34-36}. However, the reported MOFs solid acids still show constrained activities due to their either low acid concentrations or limited acid strength. The simultaneously enhance the concentrations and strength of acid sites in the MOFs solid acids, and therefore to largely improve their catalytic activities remains a challenge in the field.

Herein, we develop a variety of ionic liquids grafted MOFs (UiO-66) solid superacids by the quaternization of UiO-66 with 1,3-propane sultone, and followed by ion-exchanging with H_2SO_4 or HSO_3CF_3 . The synthesized UiO-66 solid acids were designated as UiO-66- $[\text{C}_3\text{NH}_2][\text{SO}_4\text{H}]$ and UiO-66- $[\text{C}_3\text{NH}_2][\text{SO}_3\text{CF}_3]$. Notably, these samples possess the highest acid concentrations (3.27~3.33 mmol/g) and strength among the MOFs solid acids, as determined by elemental analysis, ^{31}P solid-state NMR, Hammett indicators method and potentiometric titration. The ordered and

uniform micropores of UiO-66 solid acids result in their unique confinement effect for methanol in the transesterification of plant oil with methanol, and esterification of fatty acid with methanol, both of which are the important reactions for production of biodiesel in the industry. The superacidity together with unique confinement effect make the methanol can be precisely activated by the superacid sites located in the micropores of UiO-66, which effectively prevents the occurring of inverse reactions, and thus to accelerate the biodiesel catalytic production. As a result, the yields of biodiesel products catalyzed by UiO-66 solid acids are far better than those of Nafion NR50 superacid, Amberlyst 15 and sulfonic group functionalized ordered mesoporous silica (SBA-15-SO₃H-0.2), and their activities were even better than the homogeneous counterpart and neat H₂SO₄. As far as we know, the UiO-66 solid acids developed by us in this work are by far the most active MOFs solid acids, and have been successfully applied for catalyzing biodiesel production. Meanwhile, the relationship between structure and catalytic properties of the UiO-66 based superioracids have also been discussed. In addition to MOFs, other crystalline microporous materials such as ZIF-67 and TAPB-PDA COF could also be functionalized with sulfonic group with similar procedure, suggesting very good universality of the acidification method developed by us in this work.

RESULTS AND DISCUSSION



Scheme 1 Synthetic route of MOFs solid superacids of $\text{UiO-66-[C}_3\text{NH}_2][\text{SO}_3\text{CF}_3]$ and $\text{UiO-66-[C}_3\text{NH}_2][\text{SO}_4\text{H}]$.

For the purpose of simultaneously improving acid concentrations and strength of the MOFs solid acids, the synthesis procedure as illustrated in Scheme 1 was conducted in the present work. We take the UiO-66 with 1,3-propanesultone into reaction, which brings the sulfonic acid group onto it ($\text{UiO-66-[C}_3\text{NH}_2]^+$). Then, the ion exchanging with H_2SO_4 or HSO_3CF_3 was performed to give the UiO-66 ionized solid acids of $\text{UiO-66-[C}_3\text{NH}_2][\text{SO}_3\text{CF}_3]$ or $\text{UiO-66-[C}_3\text{NH}_2][\text{SO}_4\text{H}]$. The density of sulfonic group, acid strength and porosity of the solid acids can be easily tuned by controlling the sulfonation degree of UiO-66. Figure 1a, S1-S3 show the powder X-ray diffraction (XRD) patterns of parent UiO-66 and the acidified UiO-66 samples. Notably, the samples synthesized here show the same characteristic peaks with the simulation of UiO-66, indicating its highly crystalline network³⁷. After grafting of ionic liquids to give UiO-66 solid acids, which possess the same reflection patterns in comparison with parent UiO-66, indicating the crystallinity of UiO-66 were well maintained during the quaternization and acidic ion-exchanging process.

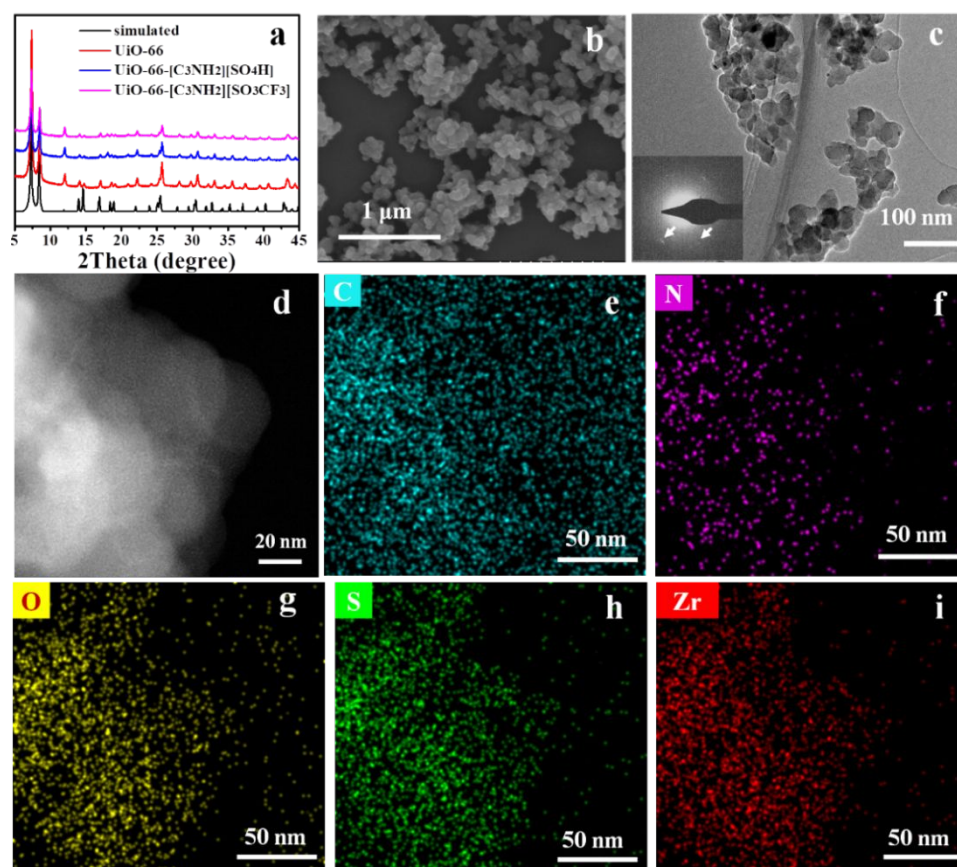


Figure 1 (a) XRD patterns, (b) SEM image, (c) TEM image, (d) HAADF-STEM, and EDS element mappings of (e) C, (f) O, (g) S, (h) N and (i) Zr of UiO-66-[C₃NH₂][SO₃CF₃].

The crystalline structures and successful functionalization with ionic liquids in UiO-66 can be further confirmed by SEM images, TEM images, EDS spectra and elemental maps. As depicted in Figure 1b-i, Figure S4 and S5, the UiO-66-[C₃NH₂][SO₃CF₃] shows uniform nanocrystalline morphology, and the diffraction spots (inside Fig. 1c) confirm its crystalline characteristics. The uniform crystalline structures of UiO-66-[C₃NH₂][SO₃CF₃] could also be observed in the STEM images (Figure 1d). In addition, the elemental maps of C, O, S, N and Zr show good dispersion in the sample, indicating the homogeneous decoration of acid ionic liquids in the UiO-66-[C₃NH₂][SO₃CF₃] (Figure 1e-i).

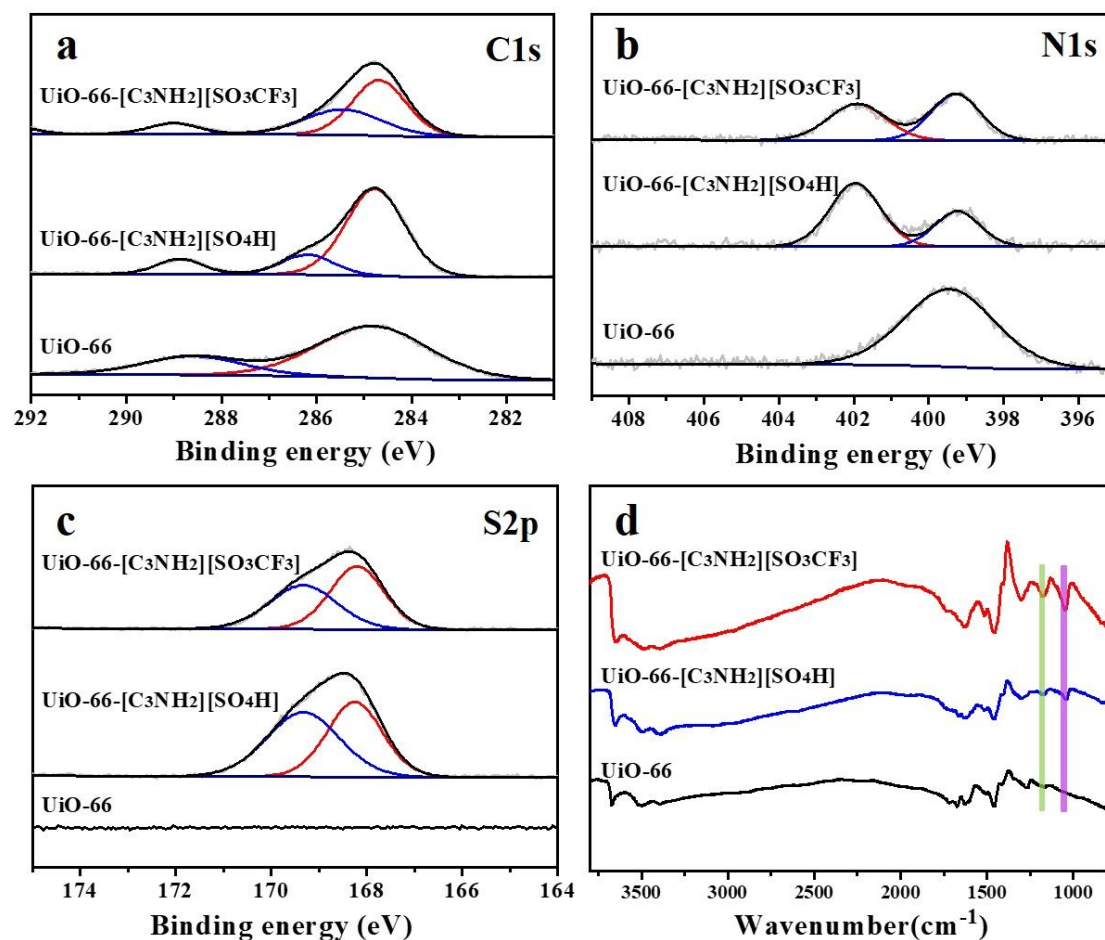


Figure 2 XPS spectra of (a) C1s (b) N1s (c) S2p and (d) FT-IR of parent UiO-66, UiO-66-[C₃NH₂][SO₄H] and UiO-66-[C₃NH₂][SO₃CF₃].

The successful grafting of strong acid ionic liquids onto UiO-66 was further investigated by XPS spectra (Figure 2, S6-S8). The parent UiO-66 shows the signals of C1s, N1s, O1s and Zr2p. After sulfonation, additional signals of S2p and F1s associated with sulfonate group, [SO₄H]⁻ and [SO₃CF₃]⁻ anions could be observed in these samples. In comparison with UiO-66, the signals of C1s in the UiO-66 solid acids can be fitted into several peaks centered at around 284.8, 286.6, 288.9 and 292.8 eV, corresponding to the C-C, C-N, C-S and C-F bonds (Figure 2), respectively^{38,39}. The signals of N1s can be fitted into two peaks, and assigned to amino group and quaternized N during sulfonation. These results confirm the successful modification

of strong acid ionic liquids onto UiO-66. Additionally, the evidence for sulfonation was obtained by analysis of UiO-66-[C₃NH₂][SO₃CF₃] and UiO-66-[C₃NH₂][SO₄H] by FT-IR spectroscopy (Figure 2d, S6-S8d, S9). In comparison with UiO-66, new bands at 1043 and 1190 cm⁻¹ associated with C-S and S=O signals could be observed in UiO-66-[C₃NH₂][SO₃CF₃] and UiO-66-[C₃NH₂][SO₄H], indicating successful grafting of acid ionic liquids onto UiO-66^{39,40}. Furthermore, UiO-66, UiO-66-[C₃NH₂][SO₄H] and UiO-66-[C₃NH₂][SO₃CF₃] show characteristic bands at around 3380~3680 cm⁻¹, which suggests the presence of -OH groups in these samples from dehydration process. As increasing the contents of acid group, the strength of the band at 1255 cm⁻¹ associate with free amine group, was significantly reduced. Moreover, the acid concentrations of the UiO-66 based solid acids (3.27~3.33 mmol/g, Table S1) were much higher than many reported MOFs solid acids.

Figure S10-S12 show N₂ isotherms of UiO-66 and its derived solid acids. The UiO-66 exhibits high volume adsorption at low pressure, indicating presence of abundant micropore in the sample, and its BET surface area was as high as 1100 m²/g. After grafting superacid ionic liquids onto its framework, the resultant solid acids show decreased BET surface areas (15-886 m²/g) and pore volumes (0.015-0.126 cm³/g), which gradually decrease as increasing the acid concentrations in the samples (Table S1). This phenomenon is reasonable since the acid groups would occupy a large portion of the micropore. However, the reduction in the surface areas caused by occupation of the pores had no negative impact on catalytic activities of UiO-66 solid acids. This is due to the high flexibility of the alkyl sulfonic group, which allows

entry and exit of small guest molecules.

Another crucial factor for catalytic application is a sufficient stability of the solid acids. To evaluate thermal stabilities of UiO-66 solid superacids, a combined TG/MS (thermogravimetric/mass spectrometry) study was performed in presence of oxygen, which was shown in Figures S13-17. The weight loss in the temperature range of 25~150 °C should be attributed to the desorption of surface vapour in parent UiO-66. The weight loss in the temperature ranges of 200~300 °C should be assigned to the desorption of the internal adsorbed water and trace CO₂ from the air because of the basic NH₂ group in the UiO-66 shows strong interactions with acid CO₂ and H₂O in the air (Figure S13). After introducing strong acid ionic liquids into UiO-66, the resultant UiO-66 solid acids show four weight loss steps at the temperature ranges of 30~200 °C, 200~300 °C, 300~420 °C and 420~580 °C, which should be attributed to the evaporation of moisture and impurities in the pores, decomposition of alkyl sulfonic group and -NH₂ group, destruction of the frameworks and collapse of internal defective structures, respectively. In addition to the TG curves, the MS technique was used to collect information during the decomposition process of the UiO-66 solid superacids. The main gaseous products were H₂O ($m/z=18$), CO₂ ($m/z=44$), NO₂ ($m/z=46$) and SO₂ ($m/z=64$) from the framework combustion. The $m/z=18$ (H₂O) fragment starts before 200 °C, which derived from physically adsorbed solvent and H₂O. The MS signals of $m/z=44$ (CO₂) and $m/z=64$ (SO₂) associated with the decomposition of alkyl sulfonic group could be nicely visualized in the temperature range of 200~420 °C during combustion of UiO-66 solid acids. The

decomposition/combustion of the organic framework begins at ca. 420 °C due to the increased intensity of the MS signal of $m/z=44$, which is characteristics of decomposition of aromatic ring and the sulfonic group (SO_2 , $m/z=64$), and oxidation of the $-\text{NH}_2$ groups at the aromatic ring. The formed NO_2 ($m/z=46$) was derived from the cracking of the organic framework of these temperatures⁴²⁻⁴⁴. Notably, as the calcination temperature was increased up to 700 °C, the content of residual ZrO_2 in the UiO-66 was nearly 41.5wt% based on the TG result, which was nearly the theoretical value (42.1wt%). After grafting superacid ionic liquid onto UiO-66 to give UiO-66- $[\text{C}_3\text{NH}_2][\text{SO}_3\text{CF}_3]$ solid superacid, which shows the ZrO_2 contents at around 28.3wt% based on TG result. In comparison with UiO-66 support, the decreased weight percentage of ZrO_2 in the UiO-66- $[\text{C}_3\text{NH}_2][\text{SO}_3\text{CF}_3]$ was due to the successful introduction of superacid ionic liquid group on to its framework (Figure S18-S20). Based on the weight changing of UiO-66 after grafting of superacid ionic liquid (13.2wt%), the sulfur content was calculated at around 3.4 mmol/g, which show relatively good consistency with that of elemental analysis results (3.3 mmol/g). The above results suggest the synthesized UiO-66 solid superacids show superior thermal stabilities.

Besides porosity and stability, the acidity of UiO-66 solid acids also plays key role for their catalytic applications. Their acid types and strengths were investigated by ^{31}P NMR techniques involving adsorbed trimethylphosphine (TMP) and trimethylphosphine oxide (TMPO) as probe molecules, which is a sensitive and reliable approach to determine the acid type (Brønsted or Lewis acid) and strength of

acid catalysts^{35,36,45-47}. As shown in Figure 3a, the ³¹P resonances at around ca. 0 ppm of adsorbed TMP were assigned to the protonated adducts of [(CH₃)₃P-H]⁺, and thus confirmed the presence of Brønsted acidity in all these MOFs solid acids^{46,47}. Besides, the ³¹P peaks at -35 and -49 ppm associated with the interaction between TMP and Lewis acid sites, were also observed for parent UiO-66 and that grafted with a small amount of RSO₃H (UiO-66-[C₃NH₂]-0.5). Therefore, both Brønsted and Lewis acid sites can be found in the UiO-66 solid acids. Thereinto, the Lewis acidity may be originated from the structural defects in the materials⁴⁸ (i.e., the terminal ZrOH groups on the node vacancies or unsaturated Zr coordination). It is noteworthy that our ³¹P-TMP NMR experimental results also indicated that the Lewis acidity of parent UiO-66 gradually vanished during the treatment with strong acidic ionic liquid (HSO₃CF₃), suggesting the acidity of parent UiO-66 could be controlled by grafting with sulfonate group.

The acid strengths of these UiO-66 solid acids were further investigated by TMPO adsorption, in which a larger ³¹P chemical shift of adsorbed TMPO usually corresponds to a stronger acidity⁴⁶. The parent UiO-66 displayed only one distinct ³¹P Resonance at around 54 ppm, which was assigned to TMPO adsorbed on the weak acid sites. After grafting with sulfonate group and superacid ionic liquid, multiple ³¹P resonance peaks centered at around 63, 66, 77 and 83 ppm were evidenced, indicating the presence of Brønsted acid sites with different acid strength in the UiO-66 solid acids (Figure 3b). Moreover, two-dimensional ¹H-³¹P heteronuclear correlation (2D HETCOR) NMR technique^{46,49} was also conducted for the purpose of probing the

spatial correlations of the ^{31}P and ^1H nucleus in the TMPO adsorbed sample. As shown in Figure 4, the correlation peaks at (67, 2.5), (77, 2.7), and (83, 2.6) ppm could be ascribed to adsorbed TMPO associated with the methyl protons. While the correlation peaks at (67, 10.7), (77, 12.8), (83, 12.4), and (88, 10.8) ppm may be ascribed to TMPO adsorbed on acidic protons (i.e., TMPO interacts with Brønsted acid sites to form TMPOH^+ complexes), which correspond to ^{31}P resonance (F2 dimension) with $\delta^{31}\text{P}$ of 67~88 ppm and ^1H resonance (F1 dimension) with $\delta^1\text{H}$ of 10.7~12.4 ppm, respectively. Since the threshold $\delta^{31}\text{P}$ value of TMPO for superacidity was demonstrated to be ca. 86 ppm, with acid strength similar to 100% H_2SO_4 ⁴⁶, it can be concluded that $\text{UiO-66-}[\text{C}_3\text{NH}_2][\text{SO}_3\text{CF}_3]$ (88 ppm) contains superacid site, which would facilitate its catalytic performance (vide infra). To the best of our knowledge, the $\text{UiO-66-}[\text{C}_3\text{NH}_2][\text{SO}_3\text{CF}_3]$ possesses the largest chemical shift of TMPO (88 ppm) so far in comparison with the reported MOFs solid acids.^{35,36}

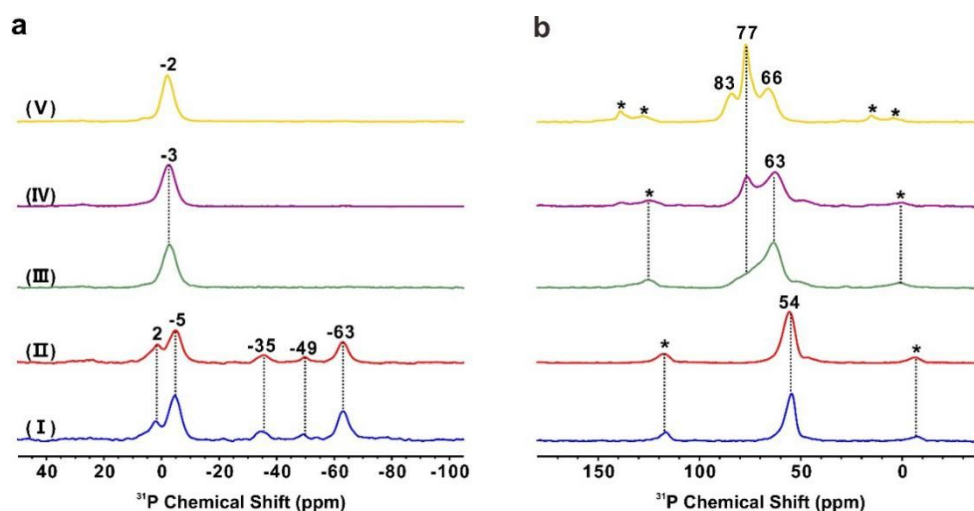


Figure 3 ^{31}P MAS NMR spectra of trimethylphosphine (TMP) (a) and trimethylphosphine oxides (TMPO) (b) adsorbed on the variety UiO-66 samples, (I) parent UiO-66, (II) $\text{UiO-66-}[\text{C}_3\text{NH}_2]\text{-0.5}$, (III) $\text{UiO-66-}[\text{C}_3\text{NH}_2]\text{-1.0}$, (IV)

UiO-66-[C₃NH₂][SO₄H] and (V) UiO-66-[C₃NH₂][SO₃CF₃].
UiO-66-[C₃NH₂][SO₃CF₃], The asterisks in the ³¹P NMR spectra denote spinning sidebands.

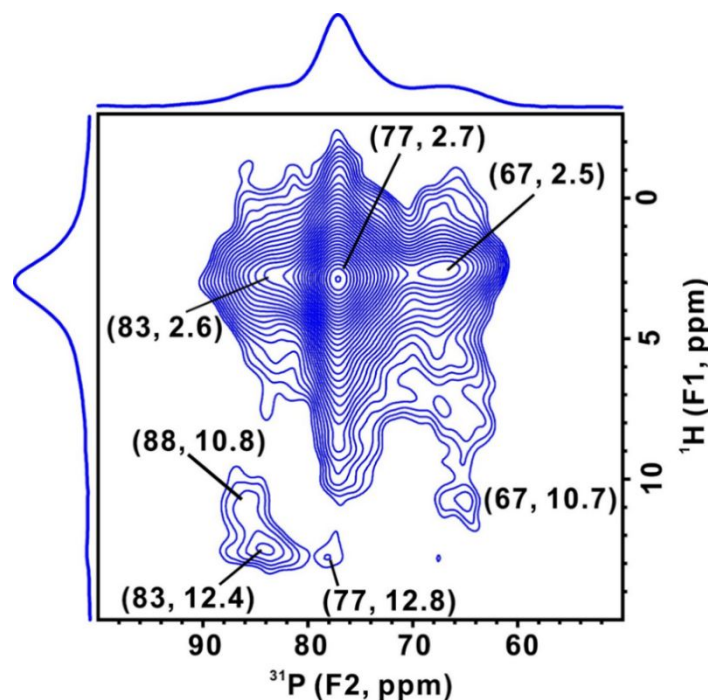


Figure 4 ³¹P-¹H Lee-Goldburg cross-polarization heteronuclear correlation (LG-CP HETCOR) of trimethylphosphine oxides (TMPO) adsorbed on UiO-66-[C₃NH₂][SO₃CF₃].

The superacidity the UiO-66-[C₃NH₂][SO₃CF₃] has been further confirmed by using Hammett indicators method (Table S2)³¹. The UV-spectra of the solutions with different indicators were collected, showing stronger adsorption peaks in UiO-66 solid acids than that of 100% H₂SO₄ (Table S2). The results confirm the acid strength of UiO-66 solid acids (H₀<-12.4) is stronger than 100% H₂SO₄³¹. Furthermore, potentiometric titration was also used to characterize the acid strength of UiO-66 solid acids, very high initial electric potential (583.1&314.8 mV) of UiO-66-[C₃NH₂][SO₃CF₃] and UiO-66-[C₃NH₂][SO₄H] further confirms their

superacidity (Figure S21)⁵⁰, in good agreement with ³¹P solid-state NMR resonance and Hammett titration results. Likely, the superacidity of UiO-66 solid acids such as UiO-66-[C₃NH₂][SO₃CF₃], should be a result of confinement effect of grafted superacid ionic liquids in its ordered micropores, which results in the much enhanced concentrations of acid sites in the microporous environment.

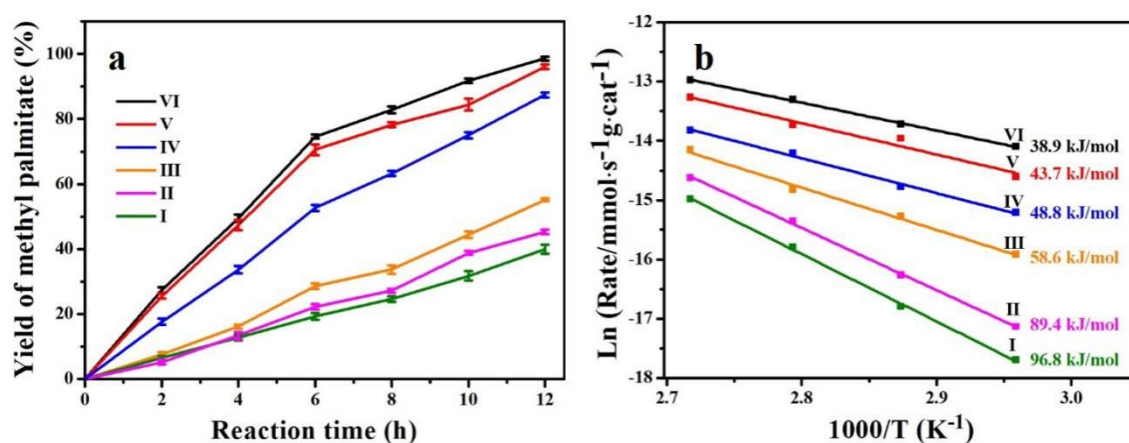


Figure 5 (a) The yields of methyl palmitate and (b) activation energies (E_a) in the transesterification of tripalmitin with methanol catalyzed by (I) SBA-15-SO₃H-0.2, (II) Amberlyst 15, (III) Nafion NR50, (IV) H₂SO₄, (V) UiO-66-[C₃NH₂][SO₄H] and (VI) UiO-66-[C₃NH₂][SO₃CF₃].

The catalytic performance of UiO-66 solid superacids were evaluated in important acid-catalyzed reactions towards transesterification and esterification to biodiesel. Figure 5 and Figure S22 depict the kinetic curves and calculated activation energy in transesterification of tripalmitin with methanol over various catalysts, in which methyl palmitate was the product. Due to superacidity, uniform and ordered microporosity, UiO-66 solid superacids show the excellent catalytic activities for transesterification of tripalmitin. The UiO-66-[C₃NH₂][SO₃CF₃] showed much better catalytic activities than those of Amberlyst-15, SBA-15-SO₃H-0.2, Nafion NR50

superacid, and its activity was even better than 100% H₂SO₄. Furthermore, UiO-66-[C₃NH₂][SO₃CF₃] and UiO-66-[C₃NH₂][SO₄H] show the activation energies of 38.9 kJ/mol and 43.7 kJ/mol, which were much lower than those of H₂SO₄ (48.8 kJ/mol), Nafion NR50 (58.6 kJ/mol), Amberlyst-15 (89.4 kJ/mol) and SBA-15-SO₃H-0.2 (96.8 kJ/mol) at the reaction conditions. Although UiO-66-[C₃NH₂][SO₃CF₃] owns the lowest BET surface area (15 m²/g), it possesses the largest chemical shift of TMPO (88 ppm) and the best activity among the UiO-66 solid acids in biodiesel catalytic production. It is concluded that the superacid sites located into the inner pores of UiO-66 play key role in the transesterification and esterification to biodiesel. In addition to purified oil, the UiO-66 solid superacids also showed impressive activities for conversion of sunflower oil to biodiesel (Table S3), there are five main products of methyl ester with varying lengths of carbon chain and unsaturated levels were detected in the reaction. The yields of biodiesel reach 86.6%~98.4% in the reactions catalyzed by the UiO-66-[C₃NH₂][SO₃CF₃], superior to UiO-66-[C₃NH₂][SO₃CF₃]-1 (84.7%~94.7%), UiO-66-[C₃NH₂][SO₃CF₃]-0.5 (78.7%~87.1%) and UiO-66-[C₃NH₂][SO₄H] (84.6%~94.4%). Their activities are much better than Amberlyst-15 (25.8%~55.7%), SBA-15-SO₃H-0.2 (48.3%~61.4%), Nafion NR50 (52.6%~62.9%) and the homogeneous acid ionic liquid of aniline-[C₃NH₂][SO₃CF₃] (80.1%~90.5%). Furthermore, the activities of UiO-66 based superacid ionic liquids are still better than 100% H₂SO₄ (82.1%~93.7%). Unlike liquid acids, the MOFs solid acids could be easily recovered by filtration. Notably, the activities of UiO-66-[C₃NH₂][SO₃CF₃] are well maintained even after

five times of cycling (85.9%~96.3%). Excellent reusability of UiO-66 solid superacids should be due to their superacid sites, abundant and ordered microporosity, which show unique confinement effect for methanol, and improve their anti carbon deposition property. Meanwhile, the acid sites in the MOFs solid acids only start to decompose at a temperature above 230 °C (Figure S14-S20), which makes the samples stable enough for recycling. Apart from transesterification, the UiO-66 solid acids were also found to be rather active in esterification of fatty acid (e.g. palmitic acid) with methanol for the biodiesel production. Their activities were again better than 100% H₂SO₄, and much better than the above mentioned solid acids (Figure S23). The located superacid sites in the uniform and ordered microporosity of UiO-66 could result in unique synergistic effect in the reactions and further accelerate the catalytic production of biodiesel.

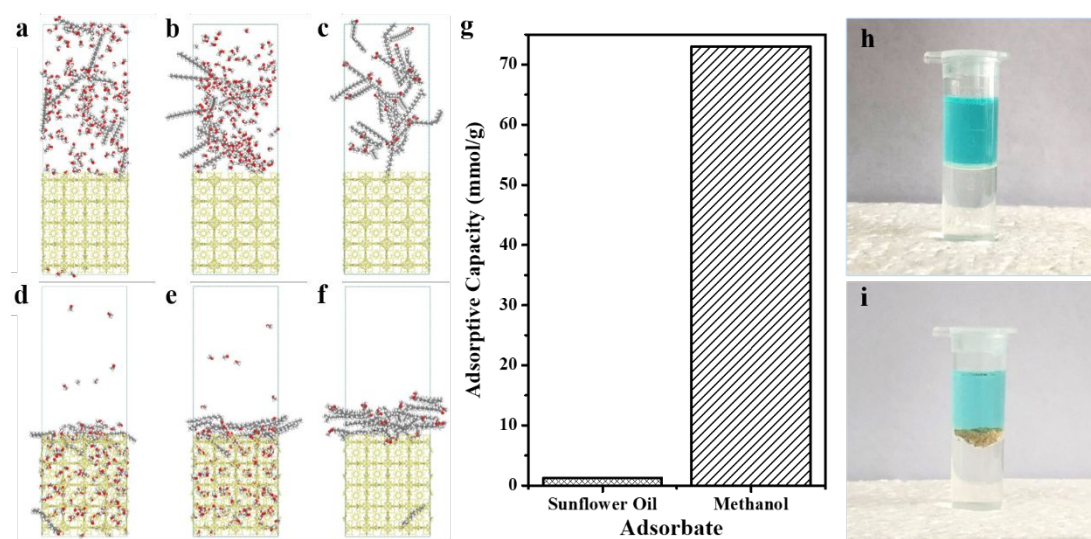


Figure 6 Initial and final structures of molecules on the UiO-66 support: (a&d) vegetable oil and methanol, (b&e) glycerol tripalmitate and methanol, (c&f) products. (g) Saturated adsorptive capacity of sunflower oil and methanol over UiO-66-[C₃NH₂][SO₃CF₃]. (h) The mixture of sunflower oil and methanol and (i) the

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4 dispersion behavior of UiO-66-[C₃NH₂][SO₃CF₃] in the mixture of sunflower oil and
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6 methanol (trace blue dye was dissolved in methanol for clarification).
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9 To better understand the relationship between structure and catalytic performance
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11 of UiO-66 solid superacids, molecular dynamic (MD) simulation was used to explore
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13 the adsorption and diffusion dynamic process of reactants and products in the
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15 reaction⁵¹⁻⁵³. As shown in Figure 6, vegetable oil was just adsorbed on the external
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17 surface of the catalysts, and most of methanol molecules were diffused into the inner
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19 channel. It's noteworthy that the larger dimension of the product will lead to its
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21 diffusion on the external surfaces. The same results of glycerol tripalmitate and
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23 products were also found (Figure 6, S24-S27). Therefore, the plausible mechanism in
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25 this reaction is that the activated methanol by the strong acid sites in the inner pores of
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27 UiO-66 solid acids, attacks to tripalmitate, and then lead to biodiesel formation. Since
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29 the comparable barriers for the inverse reaction, consequently, the inverse reaction
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31 can not usually be prevented in the homogeneous catalysts. In our case, it's apparent
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33 that almost methanol was adsorbed inside the inner channel of UiO-66 solid acids,
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35 and then was protonated by the superacid site to form activated methanol (Figure
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37 6d&e); While the tripalmitate was exclusively located on the surfaces. During the
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39 catalytic process, it's expected once the protonated methanol diffuses to micropore
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41 mouth, and was readily captured by the tripalmitate on the external surface. It's
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43 noteworthy that, the reaction between tripalmitate and activated methanol molecule
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45 were performed on the surface of UiO-66 solid acids, and thus completely avoid the
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47 restriction of mass transfer for the most reaction in porous materials (Figure 6e&f).
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Furthermore, it's observed that the quicker delivery of methanol inside inner pores, and easily compensate consumed methanol close to the surface (Figure S27), and thus reaction continues. On the other hand, the distribution of product on the external surface, thus distinctly away from the acid sites, and effectively prevent the occurring of inverse reaction as well.

The unique selective confinement of methanol can also be confirmed by the saturated adsorptive tests, the UiO-66-[C₃NH₂][SO₃CF₃] solid acid shows much higher adsorption capacity for methanol than that of sunflower oil (~58.4 times, Figure 6g). In addition, the UiO-66-[C₃NH₂][SO₃CF₃] show better wettability for methanol in comparison with sunflower oil, the UiO-66-[C₃NH₂][SO₃CF₃] solid acid was selectively dispersed onto the phase interface of the mixture of methanol and sunflower oil (Figure 6h&i). Therefore, the excellent activities of UiO-66 solid acids in the reactions of biodiesel production, were resulted from their good wettability and unique confinement activation of methanol by the superacid sites located in the inner micropores of samples. Based on both experimental and theoretical results, it can be concluded that superacid sites located in the ordered microporosity plays key factor for confinement activation of methanol, which dominates relationship between structure and catalytic performance of UiO-66 solid superacids.

In addition to MOFs support, other crystalline microporous materials such as ZIF-67 and TAPB-PDA COF could also be functionalized with acid group (e.g. –SO₃H) with similar procedures to give stable ZIF-67 and TAPB-PDA COF based solid acids. The products were analysed by XRD patterns, N₂ isotherms, XPS spectra,

FT-IR spectroscopy, and TG curves (Figure S28-35), which confirm the successful sulfonation ZIF-67 and TAPB-PDA COF with the method developed by us in this work, and the resultant solid acids show good stabilities. These results indicate the universality of the method developed by us for acidification of crystalline microporous materials, and therefore to design crystalline, stable and ordered microporous solid strong acids with superior catalytic performance in various acid-catalyzed reactions.

CONCLUSIONS

In summary, ionic liquids grafted UiO-66 solid superacids were developed by us in this work. The ionic liquid groups were homogeneously dispersed onto the UiO-66. The derived MOFs solid acids show extraordinary activities and enhanced durability for catalyzing biodiesel production because of their extremely-high acid strength, uniform and ordered microporosity. These characteristics make the methanol can be selectively confined and activated in the micropores of the samples so as to prevent occurring of inverse reaction. Their activities were even better than the homogeneous acid ionic liquid and neat H_2SO_4 , and far better than many reported solid acids. Moreover, the sulfonic group could also be grafted onto other crystalline porous materials such as COFs and ZIFs with similar procedures. This work provides a new concept for designing novel crystalline microporous solid acids with superacidity, high acid concentrations, excellent activities and good reusability, which may find useful applications for catalyzing biodiesel production in the industry.

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Notes

The authors declare no competing financial interest.

SUPPORTING INFORMATION

Supporting Information contains 3 tables and 32 figures, which includes detailed experimental details, acidity evaluation, solid-state NMR spectra, catalytic reactions, molecular dynamics simulation, and other additional characterization data.

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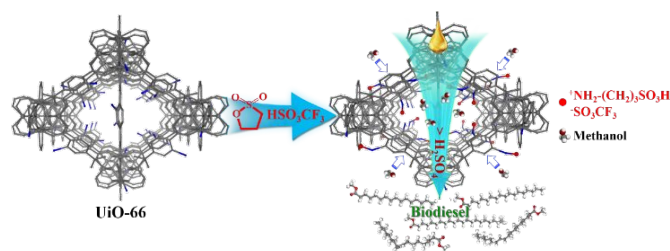
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Acceleration in biodiesel catalytic production towards transesterification was achieved by confined activation of methanol over high-concentration ionic liquids grafted UiO-66 solid superacids, which show better activities than the homogeneous counterpart and neat H_2SO_4 .